A Macroeconomic Model of the Cross-section of Currencies

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motivation

What drives the cross sectional variation of bilateral exchange rates?

International finance view: Verdelhan (18), Lustig Richmond (20), ...

- two factors account for most of variation
- countries heterogeneously exposed to factors

International macro advances: Gabaix Maggiori (15), Itskhoki Mukhin (21), ...

- models that explain exchange rate variations and connection to macro
- focus on single exchange rate, typically dollar

This paper:

- quantitative macro model that explains factor structure of exchange rates & their drivers

what we do

Macro model of the cross-section of currencies:

- TNT model of global economy with 2 ingredients:
  - heterogeneous countries: dollar assets, exposure to commodity cycle
  - segmented asset markets & global financial intermediaries
- exchange rates have factor structure: "dollar" and "commodity" factors
- dollar factor = US non-traded output + risk, commodity factor = global traded output + risk

Empirical analysis:

- dollar and commodity factors explain around 50% of variation in ΔXR
- factor loadings explained by dollar assets and export share of commodities

Quantitative analysis:

- disciplined by empirical factor structure
- **xr drivers differ by country** depending on commodity exposure and dollar assets

related literature

- **factor structure in exchange rates and currency risk**:

  Lustig Verdelhan (07), Lustig Roussanov Verdelhan (11, 14), Menkhoff Sarno Schmeling Schrimpf (12), Hassan (13), Maggiori (17), Ready Roussanov Ward (17), Verdelhan (18), Colacito Croce Gavazzoni Ward (18), Richmond (19), Lustig Richmond (20), Koijen Yogo (20), Hassan Zhang (21), ...

- **exchange rates and asset market imperfections**:

  Backus Smith (93), Lettau et al (14), Gabaix Maggiori (15), Lustig Verdelhan (19), Maggiori (22), Jiang Krishnamurthy Lustig (23), Chernov Haddad Istkhoki (24), Dao Gourinchas Itskhoki (25), ...

- **quantitative macro models of exchange rates**:

  Stockman Tesar (95), Engel West (05), Colacito Croce (11), Schmitt-Grohe Uribe (17), Itskhoki Mukhin (21, 25), Chahrour Cormun De Leo Gerron Quintana Valchev (24), Engel and Wu (24,25), Jiang Krishnamurthy Lustig (24), Kekre Lenel (24,25), Bodenstein et al (24,25), ...

Model

households' problem

Continuum of small open economies indexed by $i \in [0,1]$, household + central bank

$$\max \mathbb{E}_0 \sum_{t=0}^{t=\infty} \beta^t u\left(C_{it}\right)$$

$$P_{it}^N C_{it}^N + C_{it}^T = P_{it}^N Y_{it}^N + Y_{it}^T + R_{it} Q_{it} B_{it} - Q_{it} B_{i,t+1} + T_{it} + \Pi_{it}$$

- P_{it}^N price of non-traded good
- $Q_{it} \propto (P_{it}^N)^\alpha$ price of consumption basket $C_{it} = (C_{it}^N)^\alpha (C_{it}^T)^{1-\alpha}$
- save in local currency bond $B_{i,t+1}$ at R_{it}, get fiscal rebate T_{it}, profits of intermediaries Π_{it}

households endowments & optimality conditions

endowments

- non-tradable endowment: $Y_{it}^N = 1 + x_{it}$, idiosyncratic shocks x_{it}
- tradable endowment: $Y_{it}^T = 1 + e_i z_t$: country-specific exposure e_i to global shock z_t
- exposure to global shock $e_i \sim$ commodity incidence. Microfoundation:
  - tradable goods produced with commodities and labor
  - heterogeneous endowments of commodities across countries
  - commodity-biased productivity shocks map into z_t
  - commodity endowments of map into e_i

optimality conditions

intra-temporal condition:

$$P_{it}^N = \frac{\alpha}{1-\alpha} \frac{C_{it}^T}{C_{it}^N}$$

euler equation:

$$1 = \beta R_{it+1} \mathbb{E}_t \left[\frac{u'(C_{it+1})}{u'(C_{it})} \right]$$

central banks

Buy foreign reserves $M_{i,t+1}$ (dollar bonds), transfers net proceedings to households

$$T_{it} = R_{ut} Q_{ut} M_{it} - Q_{ut} M_{i,t+1}$$

- Q_{ut} : price of US consumption basket (dollar), R_{ut} : US interest rate

Reserve policy: $$M_{i,t+1} = M_i \left(1 + \tau \left(\frac{Q_t}{Q_{ut}} - 1\right)\right)$$

- sell reserves when dollar strong: $Q_t \equiv \int_0^1 Q_{it} \mathrm{d}i$ is a diversified currency basket
- exogenous heterogeneity in reserve level M_i
- captures domestic policy uncertainty (eg, Bianchi Hatchondo Martinez (18), Drenik Kirpalani Perez (22))

Country resource constraint:

$$C_{it}^T = Y_{it}^T + Q_{it}(R_{it} B_{it} - B_{i,t+1}) + Q_{ut}(R_{ut} M_{it} - M_{i,t+1}) + \Pi_{it}$$

global financial intermediaries

- representative intermediary. invest in currency i bonds $A_{i,t+1}$ for all i, including US
- zero-capital investment. balance sheet: $\int Q_{it} A_{i,t+1} \mathrm{d}i + Q_{ut} A_{u,t+1} = 0$

gfi problem:

$$\mathbb{E}_t \left[\underbrace{\Lambda_{t+1}}_{\text{sdf}} \int Q_{it} A_{i,t+1} \left(\underbrace{R_{i,t+1} \frac{Q_{i,t+1}}{Q_{it}} - R_{u,t+1} \frac{Q_{u,t+1}}{Q_{ut}}}_{\text{excess return vs USD}(\equiv X_{i,t+1})} \right) \mathrm{d}i \right] - \underbrace{\frac{1}{2\chi} \int \left(Q_{it} A_{i,t+1} - L_{t+1} \right)^2 di}_{\text{portfolio cost}}$$

- benchmark level $L_{t+1} = Q_{ut} \int M_{i,t+1} \mathrm{d}i$ set so intermediates absorb dollar reserves demand

optimal portfolio: $Q_{it} A_{i,t+1} = L_{t+1} + \chi \left(\mathbb{E}_t[\Lambda_{t+1}] \mathbb{E}_t[X_{i,t+1}] - \mathbb{C}_t \left[X_{i,t+1}, \Lambda_{t+1} \right] \right)$

intermediaries' pricing kernel

Intermediaries' pricing kernel is a mean-preserving spread of household pricing kernels:

$$\Lambda_{t+1} = K_t + \Gamma_t \left(\eta \int \log \Lambda^T_{i,t+1} di + (1-\eta) \log \Lambda^T_{u,t+1} \right)$$

Tradable pricing kernel captures marginal utility of tradables:

$$\Lambda^T_{i,t+1} \equiv \frac{\partial u(C_{i,t+1})}{\partial C^T_{i,t+1}}$$

- $\Gamma_t = \Gamma(1+\gamma_t)$ sets risk aversion
- K_t is a Jensen correction, ensures $\mathbb{E}_t[\Lambda_{t+1}] = \eta \int \mathbb{E}_t[\Lambda^T_{i,t+1}] di + (1-\eta)\mathbb{E}_t[\Lambda^T_{u,t+1}]$

equilibrium

Equilibrium is a set of processes $\{P^N_{it}, Q_{it}, R_{it}\}$ and $\{C^N_{it}, C^T_{it}, B_{i,t+1}, L_{i,t+1}, M_{i,t+1}\}$ such that the quantities solve the optimization problems and the following markets clear:

non-tradable goods i:	$C^N_{it} = Y^N_{it}$
tradable goods:	$\int C^T_{it} di = \int Y^T_{it} di$
currency i bonds:	$A_{i,t+1} + B_{i,t+1} = 0$
dollar bonds:	$\int \underbrace{M_{i,t+1} di}_{\text{reserves}} \underbrace{+ L_{u,t+1}}_{\text{intermediary}} \underbrace{+ B_{u,t+1}}_{\text{US HH}} = 0$

Shocks: $z_{t+1} = \rho_z z_t + \sigma_z \epsilon_{z,t+1}$, $x_{i,t+1} = \rho_x x_{it} + \sigma_x \epsilon_{i,t+1}$, and $\gamma_{t+1} = \rho_\gamma \gamma_t + \sigma_\gamma \epsilon_{\gamma,t+1}$

Solving the model

solution

Linear approximation around steady state in the limit of small shocks, large risk aversion

- simultaneously $(\sigma_z, \sigma_x, \sigma_\gamma) \longrightarrow 0$ and $\Gamma \longrightarrow \infty$ with $(\Gamma\sigma_z^2, \Gamma\sigma_x^2, \Gamma\sigma_\gamma^2) \longrightarrow (\Gamma_z, \Gamma_x, \Gamma_\gamma)$
- steady state symmetric in everything but NFA

Pricing kernel of the intermediary:

$$\lambda_{t+1} = \kappa_t - (1 + \gamma_t)\Gamma \cdot (\lambda_z \Delta z_{t+1} + \lambda_x \Delta x_{u,t+1})$$

- $\lambda_z > 0$: global marginal utility of tradables $\uparrow$ when global supply $\downarrow$
- $\lambda_x > 0$ if $\rho > 1$: US marginal utility of tradables $\uparrow$ when non-traded consumption $\downarrow$
- covariance between λ_{t+1} and exchange rates survives in the double limit

exchange rates and capital flows

Exchange rate determined by endowments, inflows $\Delta l_{i,t+1}$, and reserve accumulation $\Delta m_{i,t+1}$:

$$q_{it} = \alpha e_i z_t - \alpha x_{it} + \underbrace{\alpha \Delta l_{i,t+1}}_{\text{inflows}} - \underbrace{\alpha M_i \Delta m_{i,t+1}}_{\text{reserves}}$$

Reserve accumulation $\Delta m_{i,t+1} \equiv \Delta m_{t+1} = \tau(\Delta q_t - \Delta q_{ut})$, positive when dollar is cheap

Capital inflows:

$$l_{i,t+1} = \underbrace{\gamma_t \cdot \chi \mathbb{C}_t[\lambda_{t+1}, \Delta q_{i,t+1} - \Delta q_{u,t+1}]}_{\text{risk premium shock}} + \underbrace{L \Delta m_{t+1}}_{\text{dollar demand}} + \underbrace{\chi \mathbb{E}_t[\Delta q_{i,t+1} - \Delta q_{u,t+1} + r_{i,t+1} - r_{u,t+1}]}_{\text{expected excess returns}}$$

Euler equations

solving for exchange rates and factors

Linear equilibrium of the form

$$q_{it} = \eta_i x_{it} + \underbrace{\theta_i z_t + \xi_i x_{ut} + \mu_i \gamma_t}_{\text{global exogenous shocks}} + \underbrace{\omega_i l_{it} + \delta_i l_{ut} + \zeta_i m_t}_{\text{endogenous states}}$$

$$q_{ut} = \theta_u z_t + \xi_u x_{ut} + \mu_u \gamma_t + \omega_u l_{ut} + \zeta_u m_t$$

solving for exchange rates and factors

The limit $\chi \longrightarrow \infty$ with $\chi^{-\frac{1}{2}} M_i \longrightarrow m_i$, assume (e_i, m_i) independent, let $\rho_x = \rho_z = 1$, $\rho_\gamma = 0$

$$q_{it} = \eta_i x_{it} + \underbrace{\theta_i z_t + \xi_i x_{ut} + \mu_i \gamma_t}_{\text{global exogenous shocks}} + O(\chi^{-\frac{1}{2}})$$

$$q_{ut} = \theta_u z_t + \xi_u x_{ut} + O(\chi^{-\frac{1}{2}})$$

solving for exchange rates and factors

Compute exchange rate depreciation $\Delta s_{i,t+1} \equiv \Delta q_{u,t+1} - \Delta q_{i,t+1}$:

$$\Delta s_{i,t+1} = -\eta_i \Delta x_{i,t+1} + (\theta_u - \theta_i)\Delta z_{t+1} + (\xi_u - \xi_i)\Delta x_{u,t+1} + (\mu_u - \mu_i)\Delta \gamma_{t+1} + O(\chi^{-\frac{1}{2}})$$

solving for exchange rates and factors

Plug in the coefficients:

$$\Delta s_{i,t+1} = \alpha(e_u - e_i)\Delta z_{t+1} + \alpha(\phi\tau(m_i - l) - 1)\Delta x_{u,t+1} + \mathbb{C}_t[\lambda_{t+1}, \Delta s_{i,t+1}] \cdot \Delta\gamma_{t+1} + \alpha\Delta x_{i,t+1}$$

Relative exposure to z_t: US commodity exposure set to $e \equiv \int e_i d\mathrm{i}$

Relative exposure to x_{ut}: consider a positive shock to x_{ut}

- -1 reflects immediate USD depreciation
- $\phi\tau m_i$ reflects that reserve accumulation leads to depreciation of i
- $-\phi\tau l$ reflects that demand for dollar bonds leads to inflows to all i

solving for exchange rates and factors

Use the functional form of the pricing kernel $\lambda_{t+1} = \kappa_t - \Gamma(\lambda_z \Delta z_{t+1} + \lambda_x \Delta x_{u,t+1})$:

$$\Delta s_{i,t+1} = \alpha(e_u - e_i)\Delta z_{t+1} + \alpha(\phi\tau(m_i - l) - 1)\Delta x_{u,t+1} + \alpha \Delta x_{i,t+1}$$
$$- \mathbb{C}_t[\lambda_z \Delta z_{t+1} + \lambda_x \Delta x_{u,t+1}, \Delta s_{i,t+1}] \cdot \Gamma \Delta \gamma_{t+1}$$

solving for exchange rates and factors

Plug in the relevant components of depreciation:

$$\Delta s_{i,t+1} = \alpha(e_u - e_i)\Delta z_{t+1} + \alpha(\phi\tau(m_i - l) - 1)\Delta x_{u,t+1} + \alpha\Delta x_{i,t+1}$$
$$- \alpha(e_u - e_i) \cdot \Gamma\sigma_z^2\lambda_z\Delta\gamma_{t+1} - \alpha(\phi\tau(m_i - l) - 1) \cdot \Gamma\sigma_x^2\lambda_x\Delta\gamma_{t+1}$$

solving for exchange rates and factors

Reorganize:

$$\Delta s_{i,t+1} = \alpha(e_u - e_i)[\Delta z_{t+1} - \Gamma\sigma_z^2\lambda_z\Delta\gamma_{t+1}] + \alpha(\phi\tau(m_i - l) - 1)[\Delta x_{u,t+1} - \Gamma\sigma_x^2\lambda_x\Delta\gamma_{t+1}] + \alpha\Delta x_{i,t+1}$$

solving for exchange rates and factors

Define factors as $\text{doll}_{t+1} \equiv \int \Delta s_{i,t+1} di$ and $\text{comm}_{t+1} \equiv \int 2\Delta s_{i,t+1}(\mathbb{1}\{e_i > e\} - \mathbb{1}\{e_i < e\}) di$

$$\Delta s_{i,t+1} = \alpha(e - e_i) \underbrace{[\Delta z_{t+1} - \Gamma\sigma_z^2 \lambda_z \Delta\gamma_{t+1}]}_{=\frac{\text{comm}_{t+1}}{\alpha(e_H - e_L)}} + \alpha(1 - \phi\tau(m_i - l)) \underbrace{[\Gamma\sigma_x^2 \lambda_x \Delta\gamma_{t+1} - \Delta x_{u,t+1}]}_{=\frac{\text{doll}_{t+1}}{\alpha}} + \alpha \Delta x_{i,t+1}$$

- this uses the independence of (e_i, m_i) in cross-section and $e_u = e$

solving for exchange rates and factors

Arrive at the two-factor structure:

$$\Delta s_{i,t+1} = \frac{e_i - e}{e_H - e_L}\text{comm}_{t+1} + (1 - \phi\tau(m_i - l))\text{doll}_{t+1} + \alpha\epsilon_{i,t+1}$$

identification from factor moments

An <u>invertible</u> map $(\sigma_x^2, \sigma_z^2, (\Gamma\sigma_\gamma)^2) \mapsto (\mathbb{V}[\text{doll}_t], \mathbb{V}[\text{comm}_t], \mathbb{C}[\text{comm}_t, \text{doll}_t])$:

$$\mathbb{V}[\text{doll}_t] = \sigma_x^2 + 2\lambda_x^2 \cdot \underbrace{\sigma_x^4(\Gamma\sigma_\gamma)^2}_{\text{pricing of US-specific risk}}$$

$$\mathbb{V}[\text{comm}_t] = \sigma_z^2 + 2\lambda_z^2 \cdot \underbrace{\sigma_z^4(\Gamma\sigma_\gamma)^2}_{\text{pricing of commodity risk}}$$

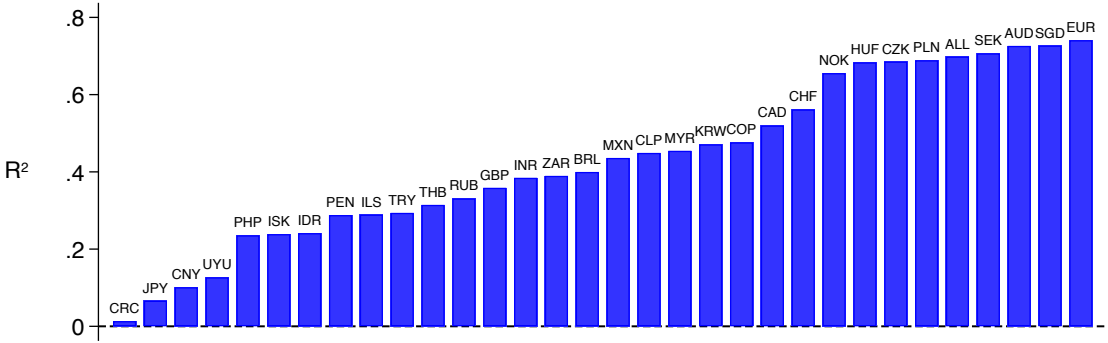
$$\mathbb{C}[\text{comm}_t, \text{doll}_t] = 4\lambda_z^2\lambda_x^2\sigma_z^2\sigma_x^2 \cdot (\Gamma\sigma_\gamma)^2$$

- both factors combine underlying shocks and <u>shocks to their pricing</u>
- factors are only correlated due to $\gamma_t \longrightarrow$ correlation informative on its strength
- variance informative on the strength of shocks to z_t and x_{ut}

Empirical analysis

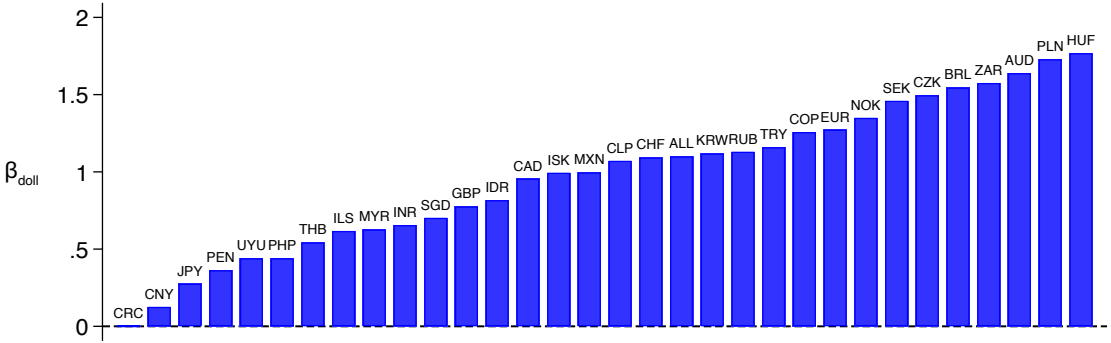
explanatory power

$$\Delta s_{i,t+1} = \beta_{i,\text{comm}}\text{comm}_{t+1} + \beta_{i,\text{doll}}\text{doll}_{t+1} + \varepsilon_{i,t+1}$$



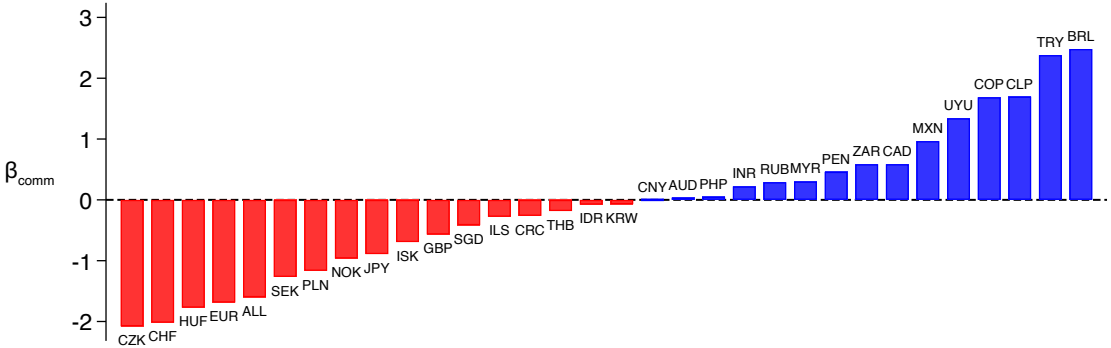
dollar loadings

$$\Delta s_{i,t+1} = \beta_{i,\text{comm}}\text{comm}_{t+1} + \beta_{i,\text{doll}}\text{doll}_{t+1} + \varepsilon_{i,t+1}$$

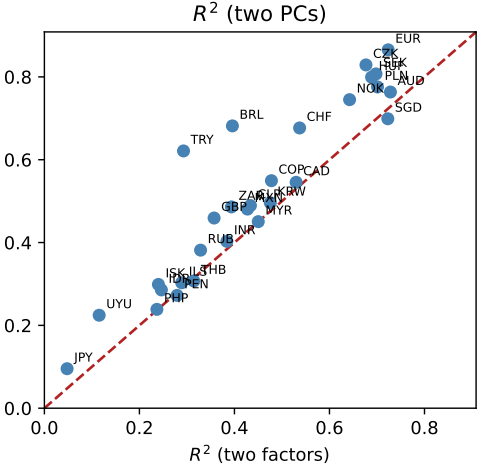


commodity loadings

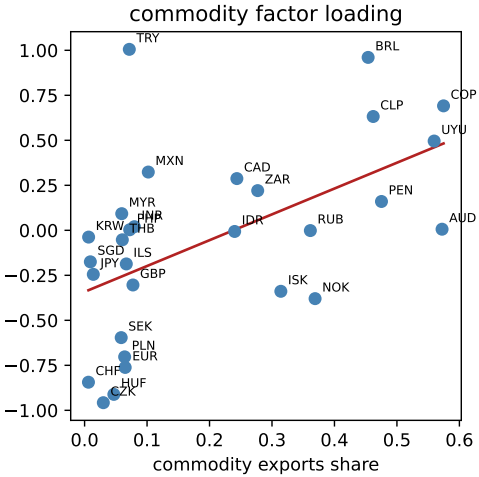
$$\Delta s_{i,t+1} = \beta_{i,\text{comm}} \text{comm}_{t+1} + \beta_{i,\text{doll}} \text{doll}_{t+1} + \varepsilon_{i,t+1}$$

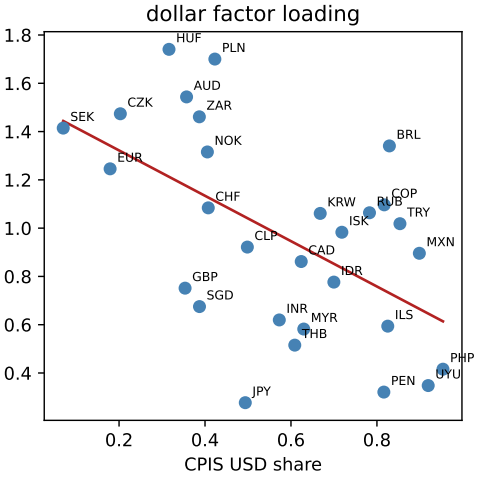


benchmarking explanatory power



determinants of loadings





questions

Do these determinants survive controlling for other cross-sectional variables?

- commodity loadings | dollar loadings

Is this about choosing USD at the base currency?

- EUR and GBP bases

How is this related to the carry-dollar regressions in the literature?

- loadings | explanatory power

Is commodity intensity related to trade centrality?

- loadings | explanatory power

Is commodity intensity related to tradable output sensitivity?

- loadings | explanatory power

Quantitative analysis

calibration

Use the three moments of factors + risk-sharing + excess volatility of exchange rates

Moment	Data	Model
$\text{std}(\text{doll}_t)$	1.94pp	1.90pp
$\text{std}(\text{comm}_t)$	1.31pp	1.30pp
$\text{corr}(\text{doll}_t,\text{comm}_t)$	0.31	0.33
$\text{corr}(\Delta s_{it}, \Delta c_{it} - \Delta c_{ut})$	-0.20	0.06
$\text{std}(\Delta s_{it})$	3.05pp	3.32pp
$\text{std}(\Delta s_{it})/\text{std}(\Delta c_{it})$	4	1
R^2 of factors	41%	50%

risk-sharing moment

parameters

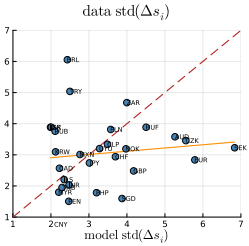
Parameter	Value	Role	Origin
α	0.43	share of non-tradables	$\text{std}(\Delta s_{it})/\text{std}(\Delta c_{it})$
L	13	total external assets	$\mathbb{C}[\Delta s_{it}, \Delta c_{it} - \Delta c_{ut}]$
σ_γ	1.29	standard deviation of innovations to γ_t	$\mathbb{C}[\text{comm}_t, \text{doll}_t]$
σ_z	3.89	standard deviation of innovations to z_t	$\mathbb{V}[\text{comm}_t]$
σ_x	3.35	standard deviation of innovations to x_{ut}	$\mathbb{V}[\text{doll}_t]$
$\hat{\sigma}_x$	4.55	standard deviation of innovations to x_{it} for $i \neq u$	R^2 of factors
τ	0.83	stabilization elasticity	reserves
ρ_z	1	persistence of z_t	tradable output
ρ_x	1	persistence of x_{it}	non-tradable output
$\{e_i, M_i\}$		distribution of country types	exports + external assets
ρ	2	inverse of EIS	
β	0.98	discount factor	
ρ_γ	0	persistence of γ_t	

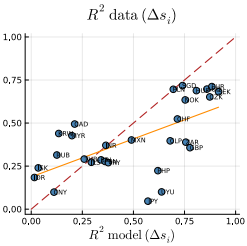
variance decomposition of US price level and factors

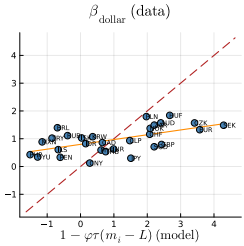
Table: variance shares

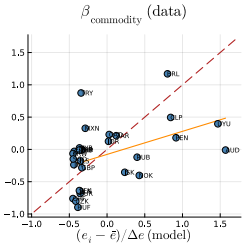
	Δq_{ut}	doll_t	comm_t
$\Delta\gamma_t$	3%	41%	26%
Δz_t	10%		74%
Δx_{ut}	86%	59%	

explaining the cross-section







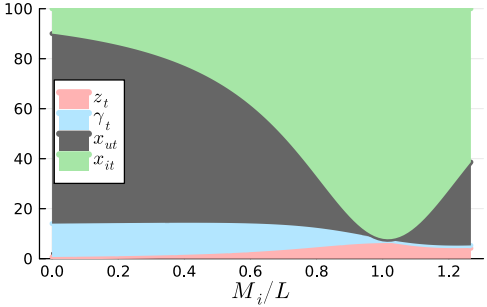


drivers of local prices

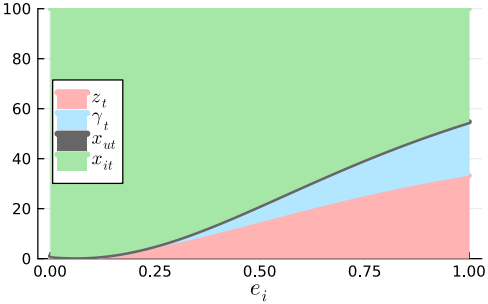
Table: variance decomposition of Δq_{it}

	Hungary	Thailand	Norway	Chile
e_i	low	low	high	very high
m_i	low	high	high	high
$\Delta\gamma_t$	9%	29%	33%	43%
Δz_t	0%	6%	19%	46%
Δx_{ut}	91%	65%	49%	11%
R^2 (model)	63%	5%	57%	54%
std (model)	3.2%	2.0%	3.0%	2.9%

drivers of local prices in cross-section



(a) shares of non-traded price variance, $e_i = e$



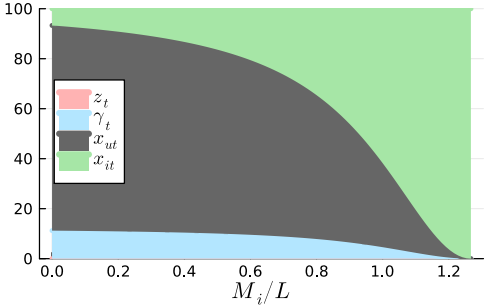
(b) shares of non-traded price variance, $M_i = L$

drivers of bilateral exchange rates

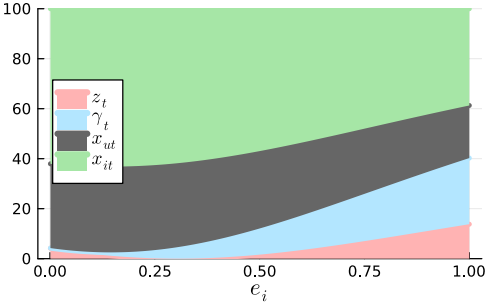
Table: variance decomposition of Δs_{it}

	Hungary	Thailand	Norway	Chile
e_i	low	low	high	very high
m_i	low	high	high	high
$\Delta\gamma_t$	7%	12%	21%	34%
Δz_t	1%	12%	2%	11%
Δx_{ut}	92%	0%	77%	56%
R^2 (data)	69%	29%	63%	40%
R^2 (model)	81%	26%	75%	68%
std (data)	3.9%	2.0%	3.2%	3.3%
std (model)	4.5%	2.3%	4.0%	3.5%

drivers of exchange rates in cross-section



(a) shares of exchange rate variance, $e_i = e$



(b) shares of exchange rate variance, $M_i = L$

conclusion

Macroeconomic model:

- output + financial shocks $\mapsto$ dollar and commodity factors
- heterogeneity in exposure to commodities and dollar assets $\mapsto$ loadings on factors

Replicate factors and loadings in the model

Decompose variation in exchange rates into fundamental sources

- financial shocks contribute <u>more</u> to dollar factor than to US prices
- financial shocks and US shocks consistently important
- high dollar assets insulate currencies from global shocks

Euler equations and Backus-Smith

Consumption and exchange rate are related:

$$c_{it} = \frac{1-\alpha}{\alpha} q_{it} + x_{it}$$

Backus-Smith covariance:

$$\mathbb{C}[c_{it} - c_{ut}, q_{it} - q_{ut}] = \frac{1-\alpha}{\alpha} \mathbb{V}[q_{it} - q_{ut}] + \mathbb{C}[q_{it} - q_{ut}, x_{it} - x_{ut}]$$

Euler equation leads to

$$r_{it} = \frac{1-\alpha}{\alpha} \mathbb{E}_t[\Delta q_{i,t+1}] + \mathbb{E}_t[\Delta x_{i,t+1}]$$

back

commodity loadings

	(1)	(2)	(3)	(4)	(5)	(6)
Commodity factor loading β_{comm}						
USD	2.64***	2.87***	2.85***	2.54***	2.56***	2.51***
	(0.68)	(0.72)	(0.72)	(0.72)	(0.72)	(0.87)
EXP	2.25***	2.80***	2.75***	2.41**	2.40**	2.67**
	(0.79)	(0.89)	(0.93)	(0.87)	(0.93)	(1.21)
R^2	0.56	0.57	0.57	0.65	0.66	0.68
Controls		(1) + cent.	(2) + open.	(3) + size	(4) + dist.	(5) + grav.
Obs.	32	32	32	32	32	32

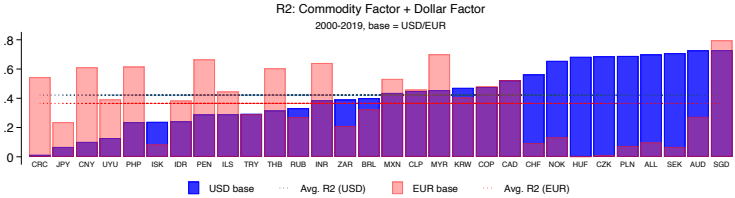
back

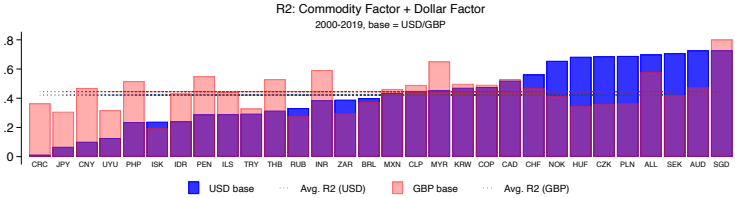
dollar loadings

	(1)	(2)	(3)	(4)	(5)	(6)
Dollar factor loading β_{doll}						
USD	-1.23^{***}	-1.27^{***}	-1.30^{***}	-1.28^{***}	-1.29^{***}	-1.30^{***}
	(0.27)	(0.29)	(0.28)	(0.29)	(0.29)	(0.31)
EXP	0.81^{**}	0.73^{*}	0.67	0.69	0.69	1.00^{*}
	(0.37)	(0.41)	(0.43)	(0.46)	(0.46)	(0.50)
R^2	0.38	0.38	0.39	0.39	0.40	0.47
Controls		(1) + cent.	(2) + open.	(3) + size	(4) + dist.	(5) + grav.
Obs.	32	32	32	32	32	32

back

explanatory power for EUR/GBP bases





back

carry explanatory power

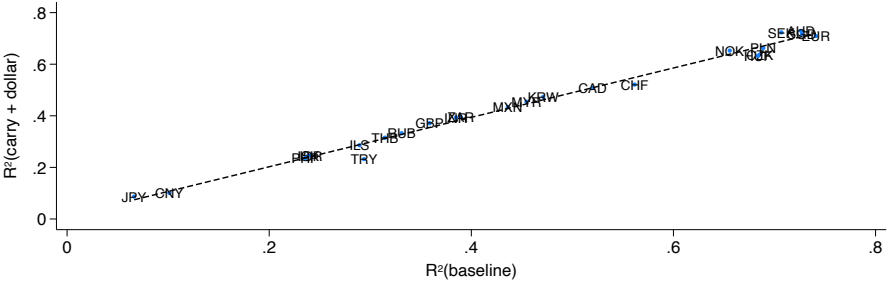


Figure: explanatory power of bivariate factor regressions with the dollar factor and either the commodity or the carry factor

back

carry loadings

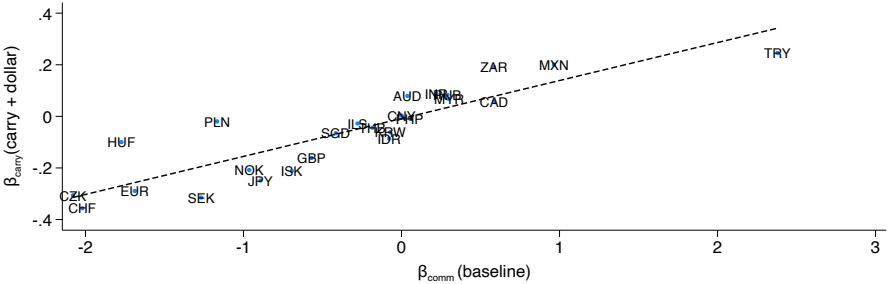


Figure: loadings on the carry factor vs commodity factor, bivariate regressions that include the dollar factor

back

trade centrality explanatory power

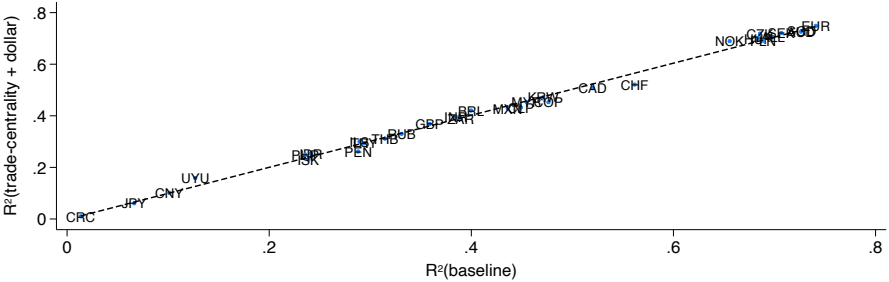


Figure: explanatory power of bivariate factor regressions with the dollar factor and either the commodity or the trade centrality factor

back

trade centrality loadings

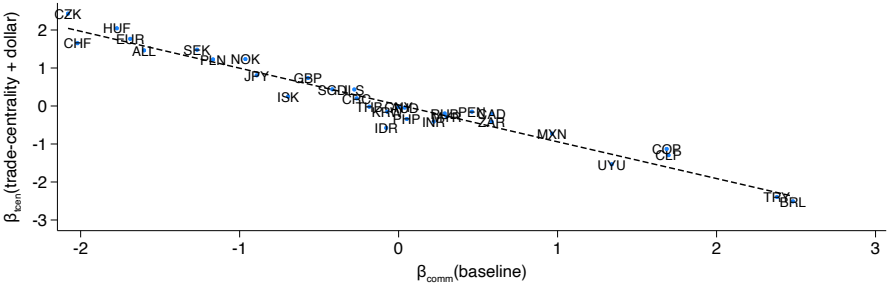


Figure: loadings on the trade centrality factor vs commodity factor, bivariate regressions that include the dollar factor

back

tradable output sensitivity explanatory power

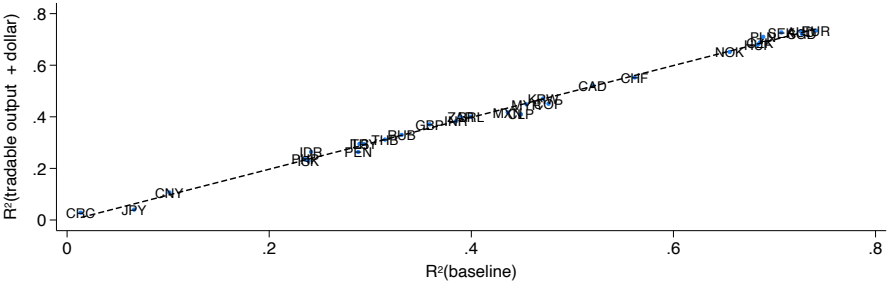


Figure: explanatory power of bivariate factor regressions with the dollar factor and either the commodity or the tradable output sensitivity factor

back

tradable output sensitivity loadings

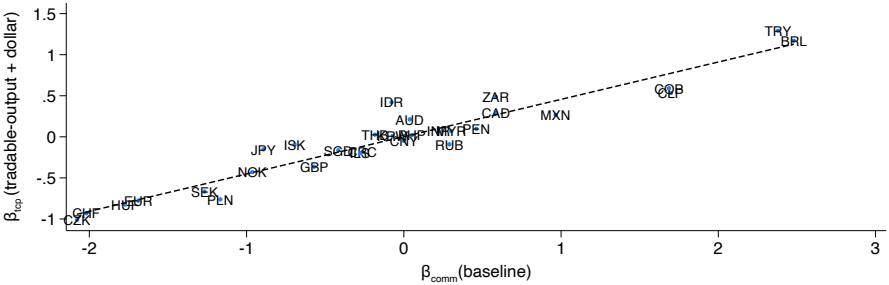


Figure: loadings on the tradable output sensitivity factor vs commodity factor, bivariate regressions that include the dollar factor

back

risk-sharing moment

back

Risk sharing typically assessed by $\text{corr}[\Delta c_{i,t+1} - \Delta c_{u,t+1}, \Delta s_{i,t+1}]$

- equal to 1 with complete and integrated markets
- close to zero (negative) in the data

Proposition 5

The risk-sharing covariance $C_i = \mathbb{C}[\Delta c_{i,t+1} - \Delta c_{u,t+1}, \Delta s_{i,t+1}]$ is

$$C_i = F_x(m_i)\sigma_x^2 - F_z(e_i)\sigma_z^2 - F_\gamma(e_i, m_i)\sigma_\gamma^2$$

Here $F_z(\cdot) > 0$ and $F_\gamma(\cdot) > 0$.

- without shocks to x_{ut}, correlation always equal to -1